MIKHIL GEORGE EDATTEL

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EDUCATION

University of Notre Dame *August 2024 - May 2028*
B.S. in Mechanical Engineering ***GPA:*** *3.73/4.00*
Concentration: Computational Engineering, Thermal & Fluid Sciences
Relevant Coursework: Thermodynamics, Solid Mechanics, Design Tools, Computational Methods, Fluid Mechanics, Differential Eqns

Honors: Certified SolidWorks Associate (CSWA) · President's Circle Mentorship Scholar · Dean's List

University of Notre Dame London - *Summer Engineering Study Abroad* *May 2025 – June 2025*

EXPERIENCE

Cooper Standard, *CAE Engineer Intern* *Northville, MI | June 2026 – August 2026*

- Achieved **10 N compression load-deflection target** for Ford F-150 Lightning outer-belt seal through iterative geometry optimization using 2D nonlinear FEA in MSC Marc/Mentat, accounting for ±2 mm door variation and +2.5 mm overslam
- **Optimized Toyota truck seal flange thickness** using FEA by comparing insertion and extraction forces across 0.95, 1.50, and 2.75 mm configurations, recommending 0.95 mm for its lower, more consistent force profile in a technical report
- Characterized installation-induced rippling in GM Silverado glass-run seals using **3D nonlinear contact FEA** in MSC Marc/Mentat, analyzing Santoprene and Celloprene deformation under installation and overcompression conditions

Domer Rover - Mars Rover Design Team, *Project Manager* *Notre Dame, IN | September 2024 – Present*

- Lead a **30+ member multidisciplinary rover team**, instituting weekly design reviews and cross-functional engineering sessions to resolve integration conflicts across mechanical, electrical, and autonomy subsystems
- Redesigned the **rover chassis** in aluminum to house electronics, computing components, and power systems while enabling interchangeable arm and science modules to be attached; validated structural performance under a **70 lb. static load**
- Diagnosed rear-biased wheel loading and front-wheel lift during **mobility testing** across sand, gravel, and rock terrains; experimentally located the rover's center of mass to redesign rocker-bogie suspension mounting points to address the issue
- Performed **dynamic load analysis and physical testing** to select an optimal rover chassis height, balancing stress on cycloidal gearbox against soil-sampling drill reach while improving **ground clearance by 25%**

RESEARCH & PROJECTS

Hypersonic Systems Lab, *Research Assistant* *Notre Dame, IN | August 2026 – Present*

- Developing an in-house **MATLAB/C++ finite-volume CFD solver** for compressible-flow and shock-wave simulations, intended to support research currently using ANSYS Fluent
- Deriving governing equations from mass, momentum, and energy conservation while investigating numerical discretization and shock-capturing methods for solver implementation

H.B. Chang Lab, *Research Assistant* *Notre Dame, IN | August 2025 – May 2026*

- Miniaturized benchtop rotary nanofabrication system (FRJS) with a 2 × 2 × 2 ft envelope into a **500 g handheld prototype** by redesigning solution delivery, airflow mechanisms, and power-system integration
- Resolved rotor instability at 10,000 RPM by redesigning the turbine-spinneret coupling in SolidWorks to correct mechanical misalignment, **validating operation across 20+ experimental runs**
- Designed an improved 6-channel annular air-jet ring to direct spun fibers toward a collection plate while reducing its overall footprint, **achieving 95% collection efficiency** in controlled deposition trials

Servo-Controlled Marble Maze *March 2026 – May 2026*

- Designed a **nested two-axis gimbal mechanism** in SolidWorks, validating independent ±20° tilt in each axis through **kinematic motion analysis** and physical prototype testing
- Produced **GD&T-based manufacturing drawings** and fabricated an electromechanical assembly using laser-cut plywood, 3D-printed components, and dual-servo actuation

SKILLS

Simulation:	MSC Marc/Mentat · Simulink · Ansys Fluent · MATLAB · Nonlinear Contact Analysis
CAD & Design:	SolidWorks (CSWA) · Siemens NX · Engineering Drawings · GD&T · Dimensional Tolerancing
Programming:	C/C++ · Python · Arduino · Numerical Methods
Manufacturing:	3D Printing · CNC · Lathe · Laser Cutting · MIG Welding · Waterjet Cutting · Soldering